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Color edge detection using the Skellam distribution as a sensor noise model

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Abstract: Conventional edge detectors suffer from inherent image noise and threshold determination. In this paper, we propose a noble edge detector based on the noise distribution for CCD or CMOS cameras. By assuming the dominant photon noise, we model the distribution of intensity differences between two neighborhood pixels. Since it is well known that photon noise follows a Poisson distribution, we introduce a Skellam distribution, which is the difference of two Poisson random variables. We show experimentally that the Skellam distribution can be used to model the noise distribution of pixels that are captured from the same scene radiance. For estimating the noise distribution given a single pixel, we find the important property that the Skellam parameters are linearly related to the intensity value of pixels. This linearity enables us to determine noise parameters according to the intensity value. In addition, parameters of the line are preserved under illumination, scene and camera setting changes except for only a gain change. Based on the noise distributions, we calculate intensity allowances of three channels for each pixel given a confidence interval. We propose a noble edge detector by skipping a pre-processing step of conventional Gaussian smoothing which is the main obstacle for robust and accurate edge detection. If the difference of intensity exceeds the intensity allowance at least in a single channel, the in-between pixel is marked as an edge pixel. We demonstrate that without conventional Gaussian smoothing the noise-model based approach can automatically extract the fine details of image structures, such as edge and corners, independent of camera setting.

Keywords: Edge detection, Noise modeling, photon noise

1. INTRODUCTION

Edge detection is finding the points in an image at which the intensity changes sharply. Sharp changes in image intensity usually reflect important details of the scene, such as discontinuities in depth and surface reflectance, changes in material properties and variation in scene illumination. For this purpose, image derivatives are computed. However, differentiation of an image is an ill-posed problem; image derivatives are sensitive to various sources of noise. Therefore, the appropriate threshold determination which classifies real edges from intensity variation caused by image noise is so difficult that ad hoc thresholds are often used in the previous works. To solve there problems, most of edge detectors reduce the image noise by Gaussian smoothing instead of modeling a noise distribution, explicitly.

Canny described what has since become one of the most widely used edge finding algorithms [1]. The first step taken is the definition of criteria which an edge detector must satisfy; high SNR (Signal to Noise Ratio), accurate localization and low false positives. The optimal filter, which is calculated by applying variational calculus to total error cost function, is shown to be a very close approximation to the first derivative of a Gaussian. Non-maximum suppression in a direction perpendicular to the edge is applied, to retain maxima in the image gradient. Finally, weak edges are removed using thresholding, which is applied with hysteresis.

Including the Canny edge detector, most approaches for edge detection smooth the image with Gaussian kernels as a preprocessing step to suppress the image [2], [3], [4], [5]. However, image smoothing also blurs sharp edges that include important information of scene details. It is the well-known problem that the derivative of a Gaussian smoothed image causes inaccurate edge localization around the corner points.

The results of the Canny edge detector [1] in MATLAB are shown in Fig. 1. Without smoothing, most false positive edges are detected given an automatic threshold because intensity variation by the image noise is not discriminated from that by the real scene structure. Even if we select the threshold manually for removing false positives, we can not remove the false edges without losing the real edges. When we apply smoothing as a preprocessing, we can reduce the false edges dramatically even by an automatic threshold. With manual smoothing scales and thresholds, we reduce the most of the false edges, however, corners of rectangles become round because of Gaussian smoothing. It means that detected edges represent uncorrect scene structure. The smoothing certainly reduce the image noise, however, the result can not preserve the actual scene structure. It directly shows fundamental problems in the conventional feature detection methods.

In this paper, we propose a noble edge detector based on the explicit noise distribution for CCD or CMOS cameras. There are several noise sources in capturing process of cameras, however, the photon noise is dominant in the sense of smallest SNR. It is well known that the photon noise follows a Poisson distribution [6], [7]. Based on the dominant photon noise assumption, we model the distribution of two neighborhood pixels as a Skellam distribu-

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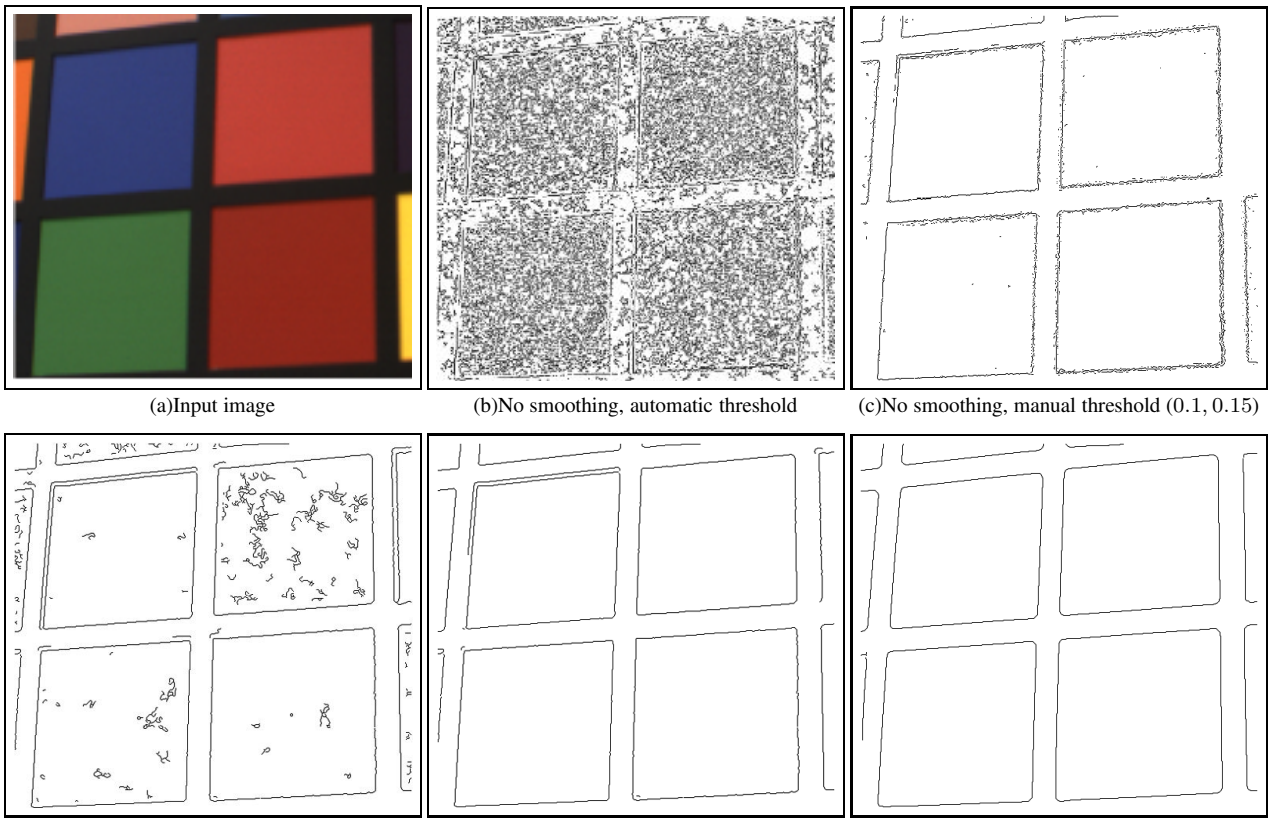


(a)Input image (b)No smoothing, automatic threshold (c)No smoothing, manual threshold $(0.1, 0.15)$

(d)Automatic smoothing, automatic threshold (e)Manual smoothing ($\sigma = 1$), manual threshold $(0.05, 0.07)$ (f)Manual smoothing ($\sigma = 3$), manual threshold $(0.03, 0.05)$

Fig. 1 Edge detection results with various parameters

tion, which is the difference between two Poisson random variables. We show experimentally that the Skellam distribution can be used to model the noise distribution of pixels of a single image at the homogeneous patch. We find that parameters of the Skellam distribution are linearly related to the intensity value of pixels, called as an Intensity-Skellam line. By this line, we can determine the explicit noise distribution for each pixel according to intensities in each channel. Importantly, the line is invariant to changes of illumination, scene and camera parameters except for the camera gain. Given a confidence interval, we calculate the intensity allowance for each pixel based on the estimated noise distribution. If the difference of intensity exceeds the intensity allowance at least in a single channel, the in-between pixel is marked as an edge pixel.

Contrary to previous works, an proposed edge detector based on our noise modeling does not smooth images because we know the reliable range of intensity variation due to sensor noise. In spite of skipping the noise reduction step (e.g. the Gaussian smoothing), the proposed edge detector not only detects fine edges corresponding to scene details, but also suppresses the unavoidable image noise effectively. More importantly, the proposed edge detector shows good response to the edges around the corner points.

The remainder of this paper is organized as follows. In Section 2, we explain our noise model based on the dominant photon noise assumption. Section 3 shows how to determine intensity allowance with the linearity between the noise parameter and intensity. In Section 4, we explain our edge detector based on the estimated noise distribution. Section 5 shows the experimental results compared in various aspects. We summarize our work in Section 6.

2. IMAGE NOISE MODEL

In [6], [7], several sources of image noise are described, but most of the sources can usually be considerably reduced by appropriate design of manufacturer. They may also be ignored, because the total SNR of a complete system is typically dominated by the smallest SNR; the most dominant noise component is photon noise in CCD or CMOS cameras. We assume that the noise is caused by the dominant photon noise.

2.1 Photon noise

The probability distribution for n_e photoelectron at the collection site (in general, in a given pixel) of observation length T seconds is known to be Poisson [6], [7]:

$$f_P(n_e \mid \rho, T) = \frac{(\rho T)^{n_e} e^{-\rho T}}{n_e!} \tag{1}$$

where ρ is the rate parameter measured in photoelectrons per second.

The average value and the standard deviation of the distribution in Eq. (1) are given by:

$$\mu = \rho T \tag{2}$$

$$\sigma = \sqrt{\rho T} \tag{3}$$

The three traditional assumptions about the relationship between signal and noise do not hold for photon noise:

- photon noise is not independent of the signal.
- photon noise is not Gaussian.
- photon noise is not additive.

The above properties of photon noise are important for the derivation of our work. The photon noise is a fundamental limitation and cannot be eliminated.

2.2 Noise modeling based on dominant photon noise assumption

As shown in Eq. (2), μ means the number of photoelectrons over an interval T. It is natural that the number of photoelectrons in a brighter pixel is greater than that in a darker pixel. From this we can predict that μ will increase as the brightness or intensity increases.

We use a Poisson distribution to represent the intensity difference distribution to avoid the erroneous Gaussian approximation. To model the noise distribution, we cannot use a Poisson distribution directly, because the photon noise is not additive. In other words, we cannot separate a noise component from the intensity of a pixel, itself. Thus, we take the difference between two corresponding pixels to remove the actual scene irradiance component. Subsequently, we can model the distribution only for noise.

The difference between two Poisson random variables is defined as a Skellam distribution [8]. The probability mass function (pmf) of a Skellam distribution is a function of k, which means the difference between two Poisson random variables. It is expressed as:

$$f_S(k; \mu^{(1)}, \mu^{(2)}) = e^{-(\mu^{(1)}+\mu^{(2)})} \left(\frac{\mu^{(1)}}{\mu^{(2)}}\right)^{k/2} I_k(2\sqrt{\mu^{(1)}\mu^{(2)}}) \tag{4}$$

where $\mu^{(1)}$ and $\mu^{(2)}$ are the means, or expected values, of the two Poisson distributions, and $I_k(z)$ is the modified Bessel function of the first kind.

2.3 Skellam parameter estimation

We can estimate the Skellam parameters of intensity difference by using the statistics of Skellam distributions. The mean μ_S and variance σ_S^2 of a Skellam distribution are given by:

$$\mu_S = \mu^{(1)} - \mu^{(2)} \tag{5}$$

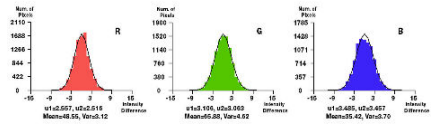
$$\sigma_S^2 = \mu^{(1)} + \mu^{(2)} \tag{6}$$

From Eq. (5) and Eq. (6), we can calculate the parameters $\mu^{(1)}$ and $\mu^{(2)}$ directly as:

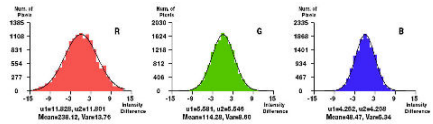
$$\mu^{(1)} = \frac{\mu_S + \sigma_S^2}{2} \tag{7}$$

$$\mu^{(2)} = \frac{-\mu_S + \sigma_S^2}{2} \tag{8}$$

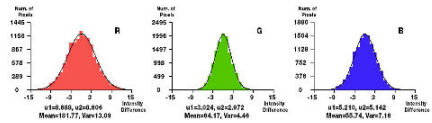
Fig. 2 Colorchecker image for estimating Skellam parameters



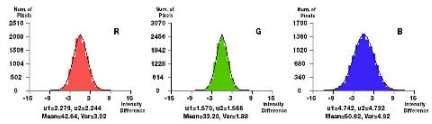
(a)Foliage patch (4th patch)



(b)Orange patch (7th patch)



(c)Moderate red patch (9th patch)



(d)Purple patch(10th patch)

Fig. 3 Skellam parameter estimation using homogeneous color patches

When we assume that all of pixels in the homogeneous regions of an image are mutually independent, we can calculate the mean and variance of a Skellam distribution as:

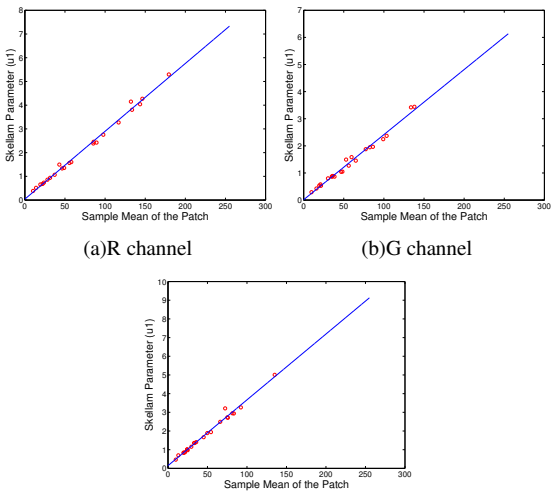
$$\mu_S = \frac{\sum_{(i,j)\in P}(x_t(i,j) - x_t(i+d_x, j+d_y))}{n} \tag{9}$$

$$\sigma_S^2 = \frac{\sum_{(i,j)\in P}(\mu_S - (x_t(i,j) - x_t(i+d_x, j+d_y)))^2}{n} \tag{10}$$

where $(i,j) \in P$ means all points in the patch and d_x, d_y are disparities in horizontal direction and in vertical direction. n is the total pixels of the patch.

For estimating the Skellam parameters, we capture the image of Gretag-Macbeth ColorChekcer as shown in Fig. 2. We slit each color patch and fit the estimated Skellam distribution from Eq. (7), Eq. (8), Eq. (9) and Eq. (10).

In Fig. 3, we show some fitting results of Skellam distributions for patches. The histograms are the real distributions of the image noise and black lines are the estimated distributions by calculating parameters. Although



(a)R channel (b)G channel

(c)B channel

Fig. 4 Linearity between sample means and the Skellam parameters

we just calculate $\mu^{(1)}$ and $\mu^{(2)}$ from patches and normalize the histogram, our fitting lines are close to the real distributions.

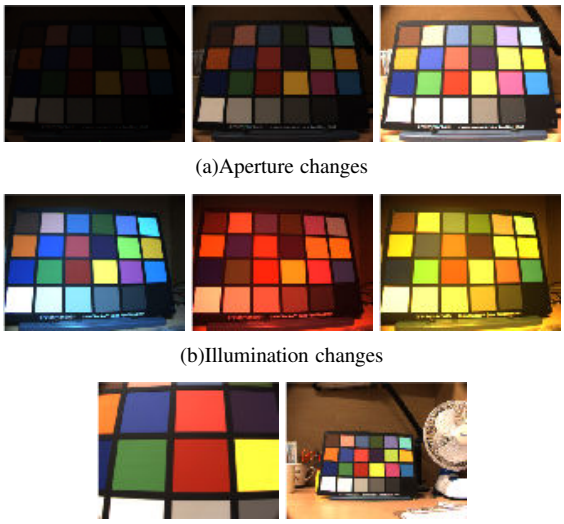
3. DETERMINATION OF INTENSITY ALLOWANCE FOR A SINGLE PIXEL

There is limitation in applying the proposed noise model to a single pixel, because a sufficient number of pixels in the homogeneous patch are necessary to estimate the Skellam parameters. We find an important property to expand our modeling scheme to a single pixel.

3.1 Intensity-Skellam line

We assume that there should be a parametric relation between Skellam parameters and intensity. From a single pixel, we cannot extract meaningful statistics for a mean or variance. To show the relationship between the intensity and the Skellam parameters, we excised 24 color patches from the image of a pattern shown in Fig. 2. We drew a scatter plot of a Skellam parameter with respect to a sample mean at each patch in Fig. 4. Fig. 4 shows the linear relationship between the sample means and Skellam parameters. We call this line an Intensity-Skellam line. We assume that a pixel value can be approximated by a sample mean and that we can estimate its Skellam parameters when we have the Intensity-Skellam lines.

Theoretically, the lines in Fig. 4 should change only with camera gain, not scene radiance or illumination because other changes expect for camera gain changes do not modify the relationship between the number of electrons and intensity. In other words, once the Intensity-Skellam line of a camera has been determined, we can use the line for images captured by the camera without modification provided the gain is fixed. To validate this fact, we captured multiple images under various changes of camera aperture, illumination and scene except for camera gain as shown in Fig. 5. We excised homogeneous



(a)Aperture changes

(b)Illumination changes

(c)Scene changes

Fig. 5 Images with various changes due to variation of camera aperture, illumination and scene

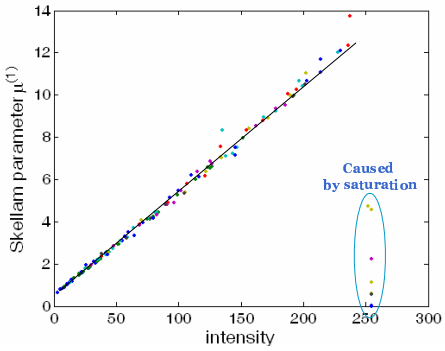


Fig. 6 Scatter plots of intensity-Skellam parameter pairs of images in Fig. 5

color patches of the colorchecker in the images of Fig. 5. Then, for each patch, we calculated a Skellam parameter and drew a scatter plot of all pairs of intensity and a Skellam parameter as shown in Fig. 6. Except for outliers caused by saturation, all of pairs are located on the same line. Therefore, the Intensity-Skellam line is preserved under changes of illumination, scene and camera parameters.

In Fig. 7, we captured multiple images, which keep the similar brightness level of the same scene, with increase of a camera gain and decrease of a camera aperture. It means that all images maintain the similar color values for corresponding pixels at each position, but captured at a different camera gain. The Intensity-Skellam lines for five images in Fig. 7 are shown in Fig. 8. Although the images have the similar view to human vision as shown in Fig. 7, noise characteristics are different according to the camera gain. Naturally, a Skellam parameter have higher slope, which means larger variation of pixel intensity, as a camera gain increases.

Fig. 7 Images with camera gain changes; camera gains increase and camera apertures decrease from the the leftmost image to the rightmost image

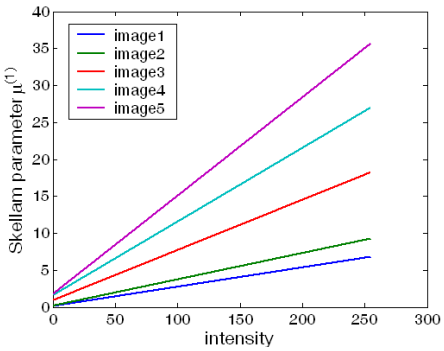


Fig. 8 Intensity-Skellam lines for images in Fig. 7

3.2 Determination of an intensity allowance

Because we have Skellam parameters for each intensity value, we have explicit noise distributions based on the Skellam parameters. Based on these distributions, we can determine an allowance for intensity variation due to the sensor noise. Our strategy for determining an intensity allowance is to test a hypothesis given a confidence interval.

In order to test a hypothesis, we must have a cumulative distribution function (cdf). Because the pmf of a Skellam distribution is defined only at integer values, we can calculate the cdf as:

$$F(K;\mu_1,\mu_2) = \sum_{k=-\infty}^{K} e^{-(\mu_1+\mu_2)} \left(\frac{\mu_1}{\mu_2}\right)^{k/2} I_k(2\sqrt{\mu_1\mu_2}) \tag{11}$$

The acceptance region for a critical value I, is:

$$A(I) = \{v|v < I\} = F(I;\mu_1,\mu_2) - F(-I;\mu_1,\mu_2). \tag{12}$$

We determine the intensity allowance I_A, that is the critical value, as:

$$I_A = \arg\max_I A(I) \quad \text{s.t. A(I)} \leq 1-\alpha \tag{13}$$

where α is the size of the type I error, which is a true rejection rate with a $(1-\alpha)*100\%$ confidence interval [9].

4. EDGE DETECTION

Our edge detection is simple because we have the significant intensity allowance based on noise estimation using a Skellam distribution. The strategy for our edge detection is very similar to that of a change detection with two consecutive images. If the difference between two neighboring pixels is within the intensity allowance, there is no edge; otherwise, there must be a real change of scene colors, which means an edge. The edge measure in both the horizontal and vertical directions is defined as:

$$e_x^c(i,j) = \frac{|I^c(i-1,j) - I^c(i+1,j)| - I_A}{I_A} \tag{14}$$

$$e_y^c(i,j) = \frac{|I^c(i,j-1) - I^c(i,j+1)| - I_A}{I_A} \tag{15}$$

where I_A denotes the intensity allowance at the pixel and c is the color channels, which are r, g, and b in RGB color space. The edge measure measures the normalized distance of intensity from the intensity allowance. If all horizontal and vertical edge measures are negative, that pixel is regarded as a nonedge pixel. Otherwise, we sum all the horizontal and vertical edge measures. Consequently, the total edge measure of a pixel is:

$$e(i,j) = \begin{cases} 0, & \text{if } \begin{cases} e_x^c(i,j) < 0 \\ e_y^c(i,j) < 0 \end{cases} \text{for all } c \\ \sum_c e_x^c(i,j) + \sum_c e_y^c(i,j), & \text{otherwise} \end{cases} \tag{16}$$

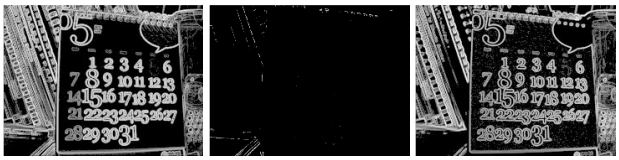
This edge measure preemptively suppresses the edge measure due to sensor noise.

When the edge measures for all pixels are calculated, we apply nonmaximum suppression because real edges do not correspond to a single line. In [1], two values are required for a hysteresis thresholding to link edges after nonmaximum suppression. However, we regard the pixels that have nonzero edge measures as being edge pixels because the intensity difference exceeds the intensity allowance determined from the noise distributions.

5. EXPERIMENTAL RESULTS

Our algorithm runs by 4Hz in AMD Athlon64 X2 2.41GHz and 2GB memory for 640x480 image size. Fig. 9 shows the normalized edge measure in various illumination conditions. The normalized edge measure is the initial step of edge detection followed by further processing, such as non-maximum suppression and hysteresis linking. Therefore, the correctness of the normalized edge measure is very crucial for edge detection. We compared the results using the proposed method with 99% confidence interval and those from the Canny operator with low and high thresholds, which are fixed for all input images. The results from the Canny operator not only cannot reduce the false edges caused by image noise, but also cannot detect find details as shown in the dark illuminated image (second input image). Our proposed method outperforms the results from the Canny operator. Without any parameter changes, our method reduce the noise correctly and detect edges even in the very dark environment. In addition, the detection of the red number 5 is difficult using the Canny operator, whereas our method detect it with the almost same amount of edge measures

(a)Input images



(b)Canny Edge (Low threshold)

(c)Canny Edge (High threshold)

(d)Proposed method (99% confidence interval)

Fig. 9 Edge detection results in various illumination conditions

of other numbers because our method combines the color information effectively.

We show edge detection results in Fig. 10 and Fig. 11. Our proposed edge detector has only one parameter, that is a confidence interval. We compared our results with those of Canny edge detector [1] in MATLAB. A Canny edge detector has three parameters: a scale of Gaussian smoothing, a low and a high thresholds for hysteresis linking. Determining the values is case-sensitive and critical to the performance. In Fig. 10, we compared the results using the proposed method and the three outputs of the Canny edge detector with an automatic, a low and a high threshold when we fixed the scale of Gaussian smoothing. Our proposed method outperforms the three Canny edge results. It suppresses the most of the false edges shown in the Canny's result with the low or the automatic threshold. Furthermore, our result shows the fine details of edges similar to the result with the low threshold. It is possible because our threshold is separately determined for each pixel by the estimated noise distribution.

In Fig. 11, we compared the results using the proposed method and the various outputs from the Canny edge detector in MATLAB with various scales and thresholds. Our proposed method outperforms the Canny edge results. It suppresses most of the false edges shown in Canny's results with the medium or the automatic threshold. Furthermore, our result shows the fine details of the edges. The common problem of conventional edges is inaccurate localization around the corner because of smoothing. As shown in the magnification view of the corner, the localization of edges around the corner using our method is well-preserved, whereas it was inferior for the Canny edge detector with a scale of smoothing sufficiently high to suppress all the unexpected edges due to image noise.

6. CONCLUSION

We propose a noble edge detection method using the noise distribution explicitly. The photon noise follows the Poisson distribution. Based on the dominant photon noise assumption, the image noise can be modeled by a Skellam distribution, which is the difference between two Poisson random variables. Skellam parameters are estimated by the mean and variance of the Skellam distribution in the homogeneous patches. The estimation results are fitted to the real noise distributions accurately. To expand the noise estimation into a single pixel case, we find the important property that the linearity between intensity and noise parameters, which is called as an Intensity-Skellam line. By showing that the line is invariant to changes in scene, illumination and camera parameters except for camera gain, we estimate the noise distributions directly with the fixed camera gain. We define the edge measure based on the intensity allowance. If the difference of intensity exceeds the intensity allowance at least in a single channel, the in-between pixel is marked as an edge pixel. Experiments show that the proposed method can detect fine details of the scene structure, as well as suppress the image noise effectively. Since our edge detection does not require any kind of smoothing process, the response around the corner points is preserved well compared with conventional edge detectors.

ACKNOWLEDGEMENT

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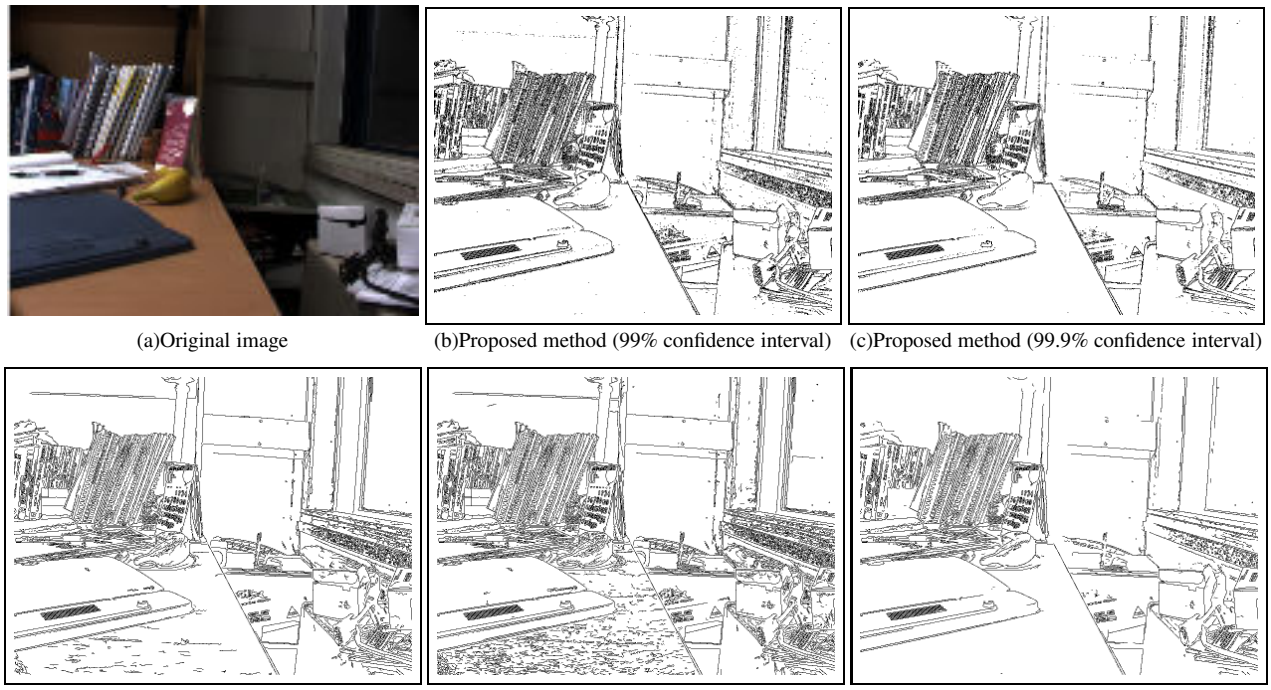

(a)Original image (b)Proposed method (99% confidence interval) (c)Proposed method (99.9% confidence interval)

(d)Canny Edge (Automatic threshold) (e)Canny Edge (Low threshold) (f)Canny Edge (High threshold)

Fig. 10 Edge detection results compared with Canny edge using various thresholds

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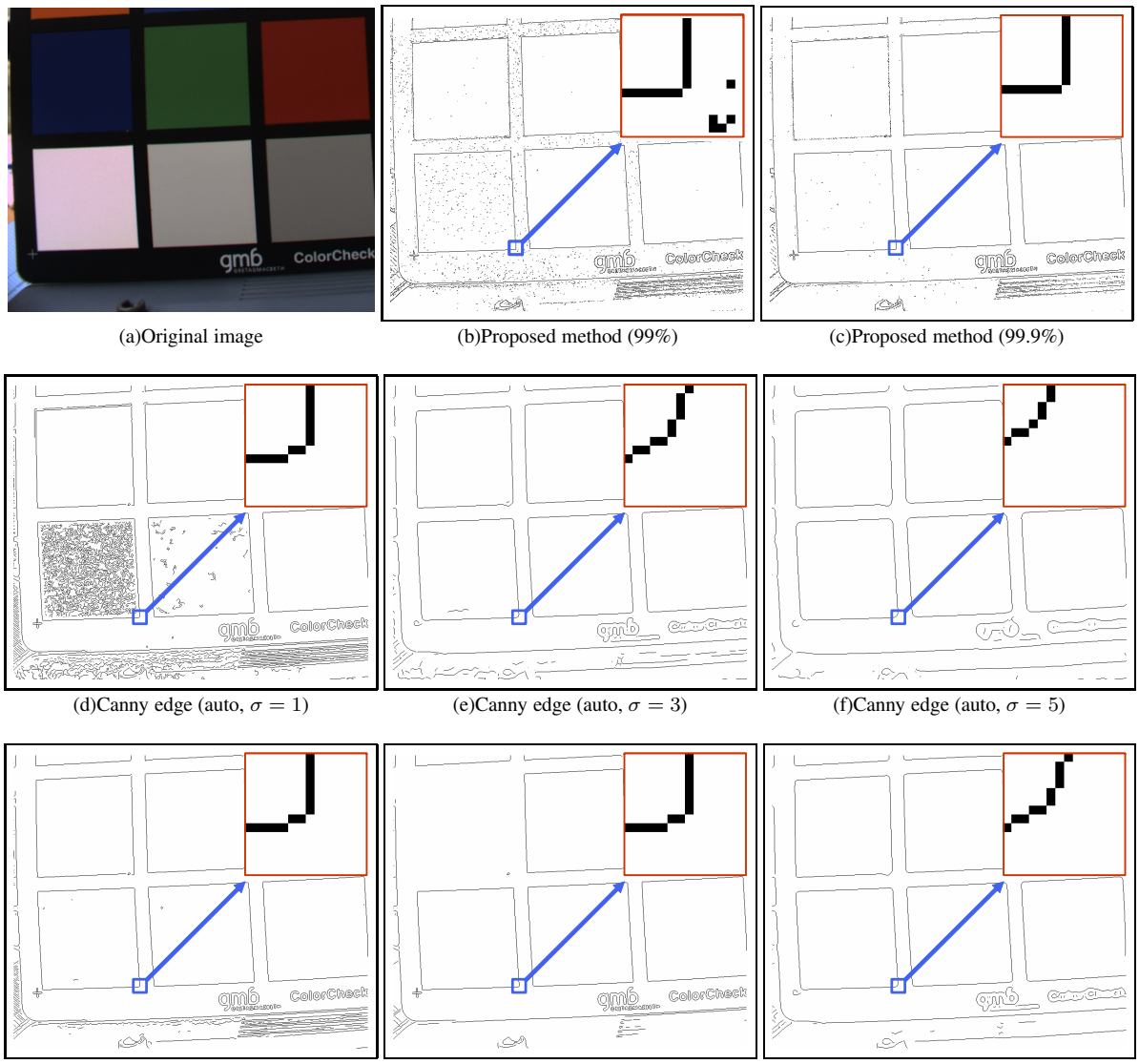

(a)Original image (b)Proposed method (99%) (c)Proposed method (99.9%)

(d)Canny edge (auto, $\sigma = 1$) (e)Canny edge (auto, $\sigma = 3$) (f)Canny edge (auto, $\sigma = 5$)

(g)Canny edge (medium, $\sigma = 1$) (h)Canny edge (high, $\sigma = 1$) (i)Canny edge (medium, $\sigma = 3$)

Fig. 11 Edge detection results compared with Canny edge using parameter settings; automatic threshold=(0.0063, 0.0156), medium threshold=(0.025, 0.035) and high threshold=(0.065, 0.075) in normalized intensity

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